

ECA5612

Ex.No:1: Simulation of FDMA in MATLAB Date

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users

and determine the channel bandwidth assigned to each user

code:

```
clc;
clear;
close all;
% Parameters
fs = 1000;          % Sampling frequency
t = 0:1/fs:1;       % Time vector (1 second)
TBW = 600;          % Total Bandwidth (kHz)
Users = 3;           % Number of users
CBW = TBW / Users;   % Bandwidth per user
% Carrier frequencies
fc = [100 300 500];
% Message signal
m = sin(2*pi*10*t);
% FDMA signals
s1 = m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t);
s3 = m .* cos(2*pi*fc(3)*t);
% Plot transmitted signals
subplot(2,1,1);
plot(t, s1, 'r', t, s2, 'g', t, s3, 'b');
grid on;
title('FDMA Channel Allocation');
xlabel('Time (s)');
ylabel('Amplitude');
legend('User 1','User 2','User 3');

% Plot channel bandwidth allocation
```

```

subplot(2,1,2);

bar(1:Users, CBW*ones(1,Users));

grid on;

title('Channel Bandwidth per User');

xlabel('Users');

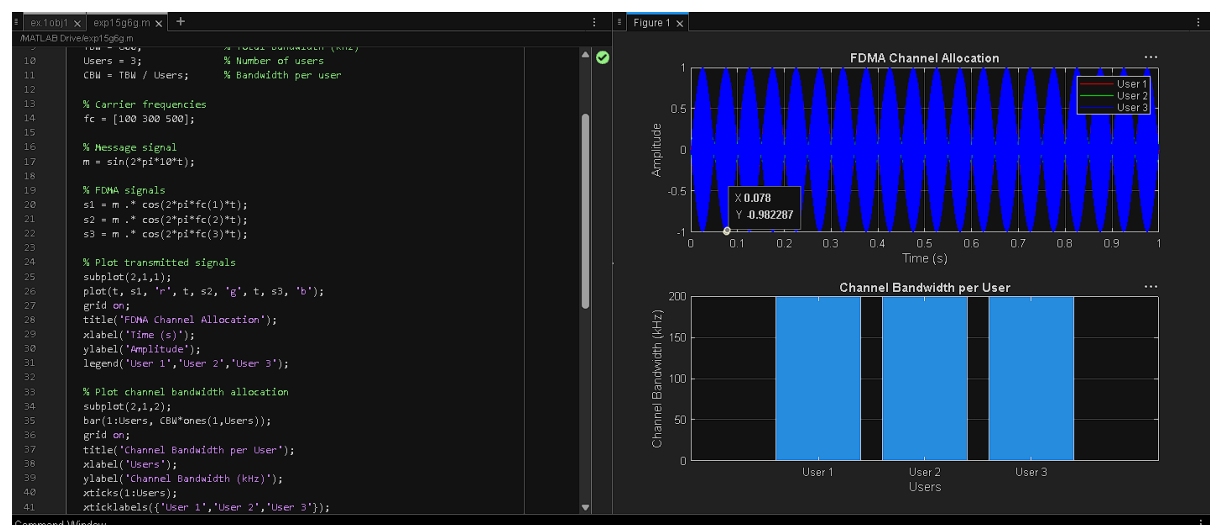
ylabel('Channel Bandwidth (kHz)');

xticks(1:Users);

xticklabels({'User 1','User 2','User 3'});

```

output:



Objective 2

Code:

```

clc; clear; close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;

M = 2; % BPSK Modulation

DataRate = BW*log2(M); % Data Rate (kbps)

Rate = DataRate*ones(1,Users);

disp(' User Data Rate (kbps)')

disp([(1:Users)' Rate'])

figure

bar(1:Users,Rate)

xlabel('Users')

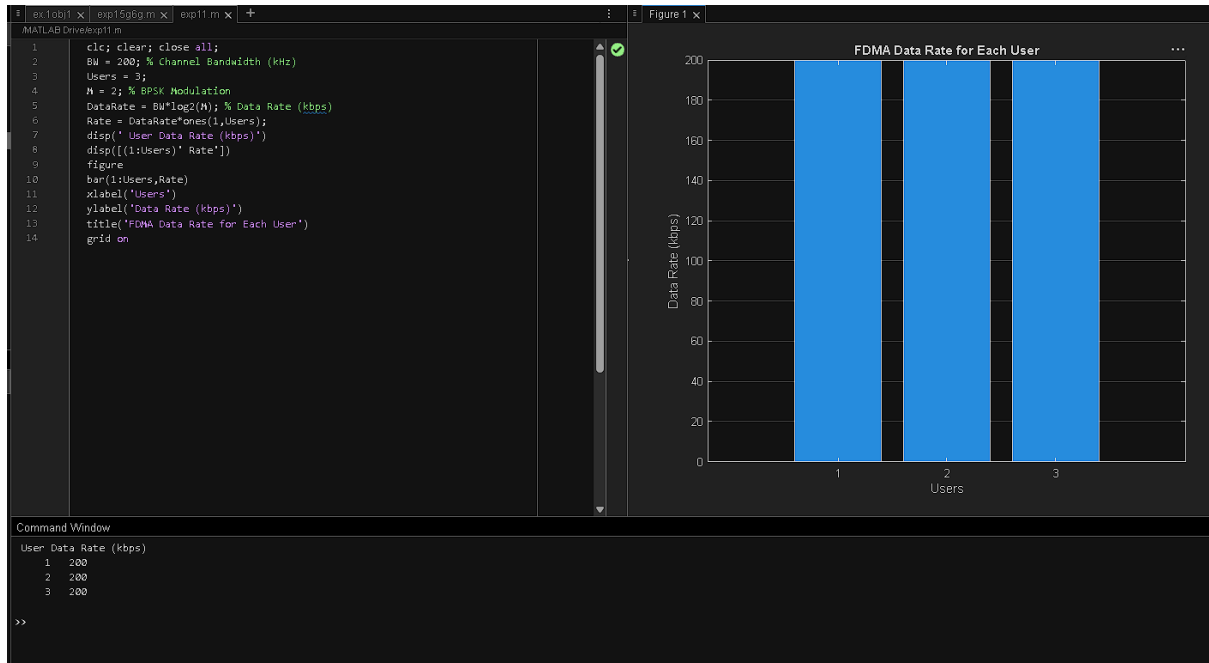
ylabel('Data Rate (kbps)')

```

title('FDMA Data Rate for Each User')

grid on

output:



Objective 3:

clc; clear; close all;

Ps = 1; % Signal Power (W)

Pn = [0.1 0.05 0.01]; % Noise Power (W)

Users = 1:3;

SNR = 10*log10(Ps./Pn); % SNR in dB

disp('User Noise Power(W) SNR(dB)')

disp([Users' Pn' SNR'])

figure

bar(Users,SNR)

xlabel('Users')

ylabel('SNR (dB)')

title('Signal-to-Noise Ratio of FDMA Users')

grid on

